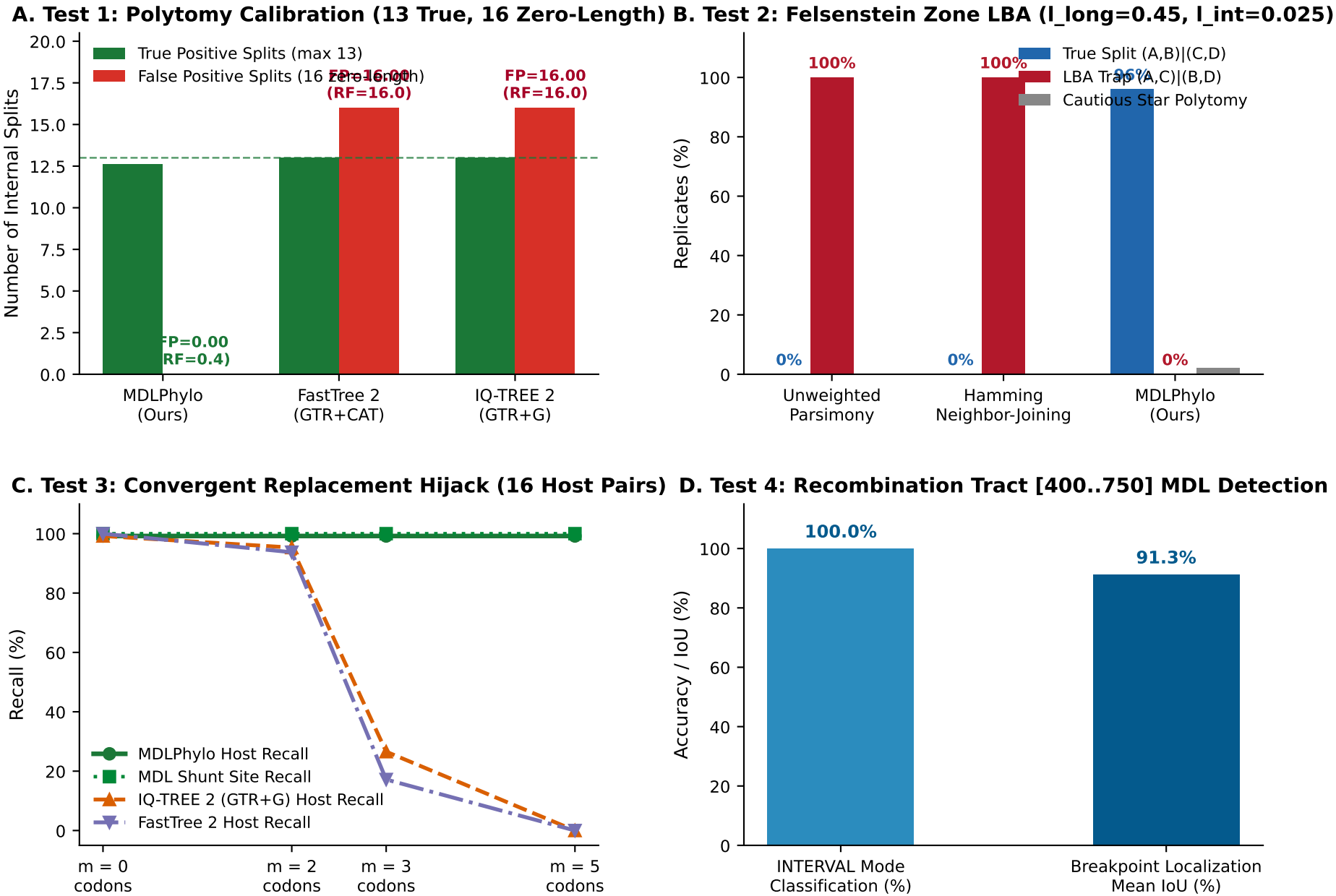
A. Test 1: Polytomy Calibration (13 True, 16 Zero-Length)
True Positive Splits (max 13)
False Positive Splits (16 zero-length)
Number of Internal Splits
FP=16.00
(RF=16.0)
FP=16.00
(RF=16.0)
FP=0.00
(RF=0.4)
MDLPhylo
(Ours)
FastTree 2
(GTR+CAT)
IQ-TREE 2
(GTR+G)
B. Test 2: Felsenstein Zone LBA (l_long=0.45, l_int=0.025)
True Split (A,B)|(C,D)
LBA Trap (A,C)|(B,D)
Cautious Star Polytomy
Replicates (%)
100%
100%
96%
0%
0%
0%
Unweighted
Parsimony
Hamming
Neighbor-Joining
MDLPhylo
(Ours)
C. Test 3: Convergent Replacement Hijack (16 Host Pairs)
Recall (%)
MDLPhylo Host Recall
MDL Shunt Site Recall
IQ-TREE 2 (GTR+G) Host Recall
FastTree 2 Host Recall
m = 0
codons
m = 2
codons
m = 3
codons
m = 5
codons
D. Test 4: Recombination Tract [400..750] MDL Detection
Accuracy / IoU (%)
100.0%
91.3%
INTERVAL Mode
Classification (%)
Breakpoint Localization
Mean IoU (%)